Mesh Network Performance Study

Packet Size: 10KB

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1. Executive Summary

This report presents the performance analysis of three different mesh network configurations with varying numbers of mobile STAs. All tests were conducted with a packet size of 10KB. The study evaluates three commercial mesh AP devices in a 400m × 400m × 30m environment with obstacles (buildings) to simulate realistic urban conditions.

Parameter	Value
Packet Size	10KB (10240 bytes)
STA Counts Tested	3, 5, 7, 9
Mesh Configurations	3 (TP-Link, Orbi, ZenWiFi)
Coverage Area	400m × 400m × 30m
Node Spacing	200m
Number of Mesh Nodes	9 (3×3 grid)
Buildings (Obstacles)	4 (15m tall, concrete)

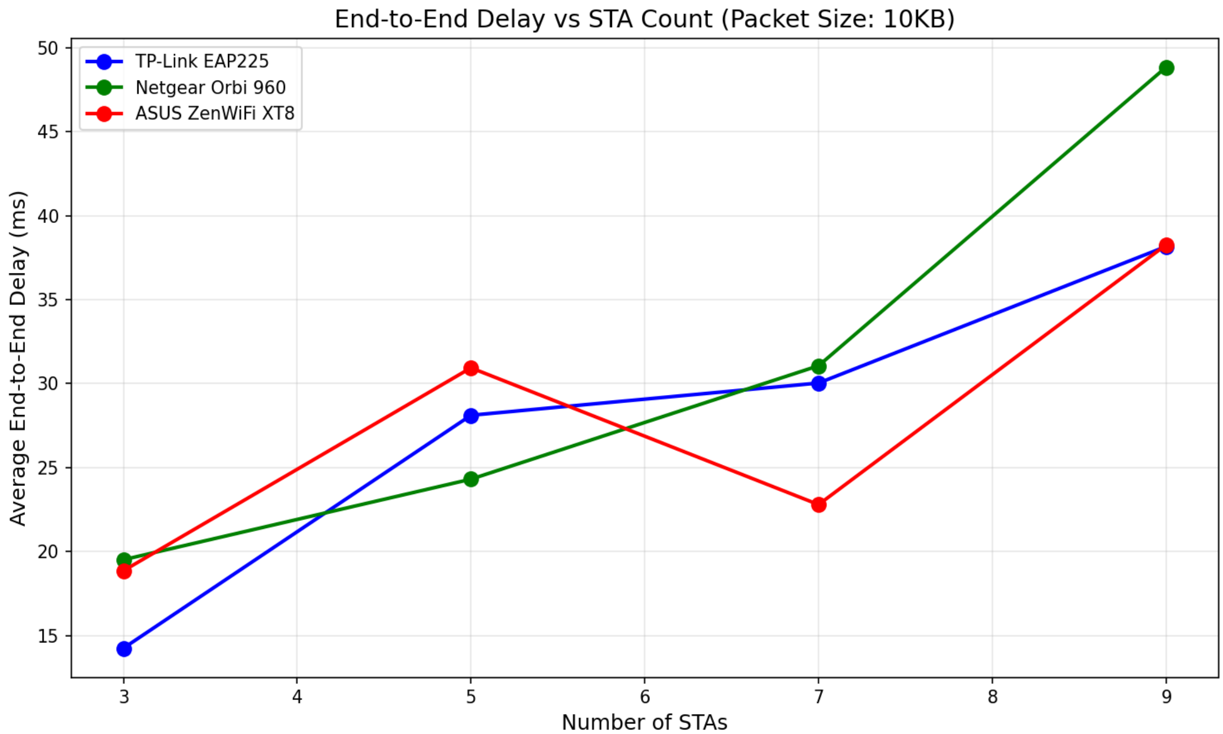
2. Network Topology

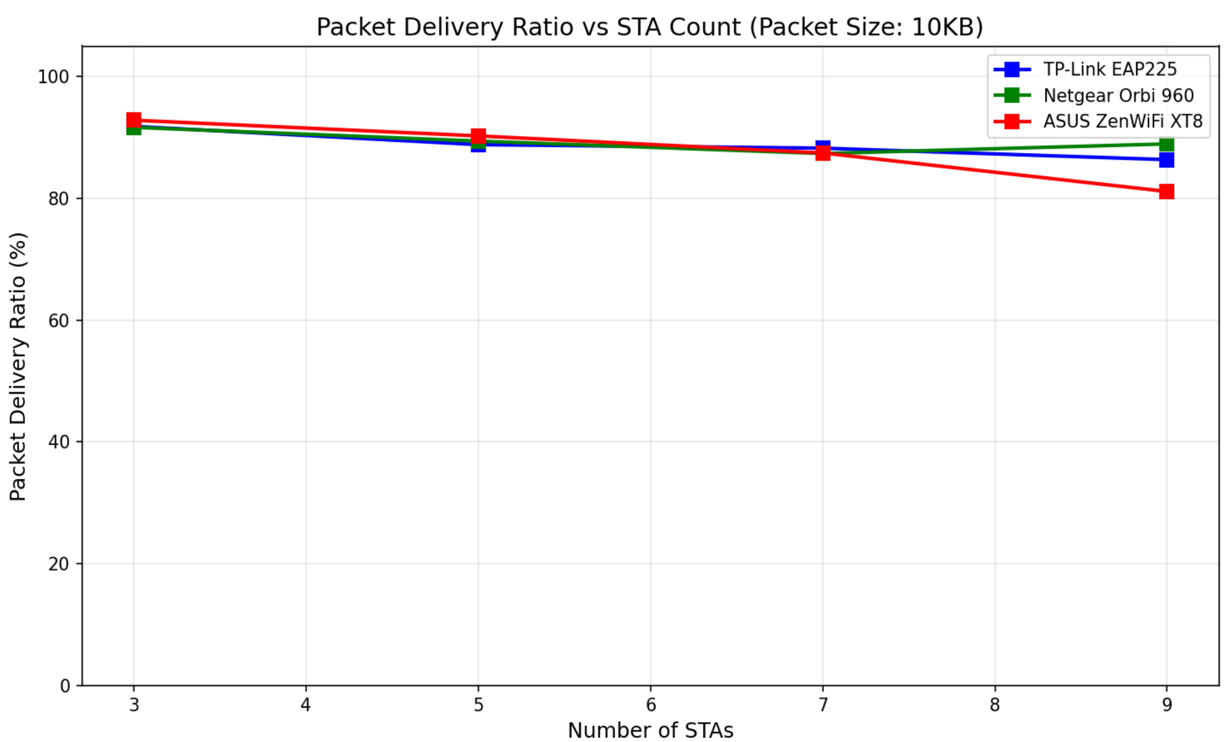
Network Topology (400m × 400m × 30m) == 0 --------- 1 --------- 2 | | | | [B3] | [B1] | | | | 3 --------- 4 --------- 5 | | | | [B4] | [B2] | | | | 6 --------- 7 --------- 8 (AP Node with STAs) | | | Gateway (STAs: 3D mobile clients) Legend: - Nodes 0-8: Mesh APs (802.11s mesh backhaul) - Node 0: Gateway to Internet - Node 8: AP serving mobile STAs (802.11ac hotspot) - [B1-B4]: Buildings (15m tall, concrete with windows) - Node spacing: 200m - Total coverage: 400m × 400m × 30m (vertical) - STAs: Mobile clients in 20m × 20m area around Node 8 - Mobility: GaussMarkov 3D (0.3-0.8 m/s, 0-30m height) ==

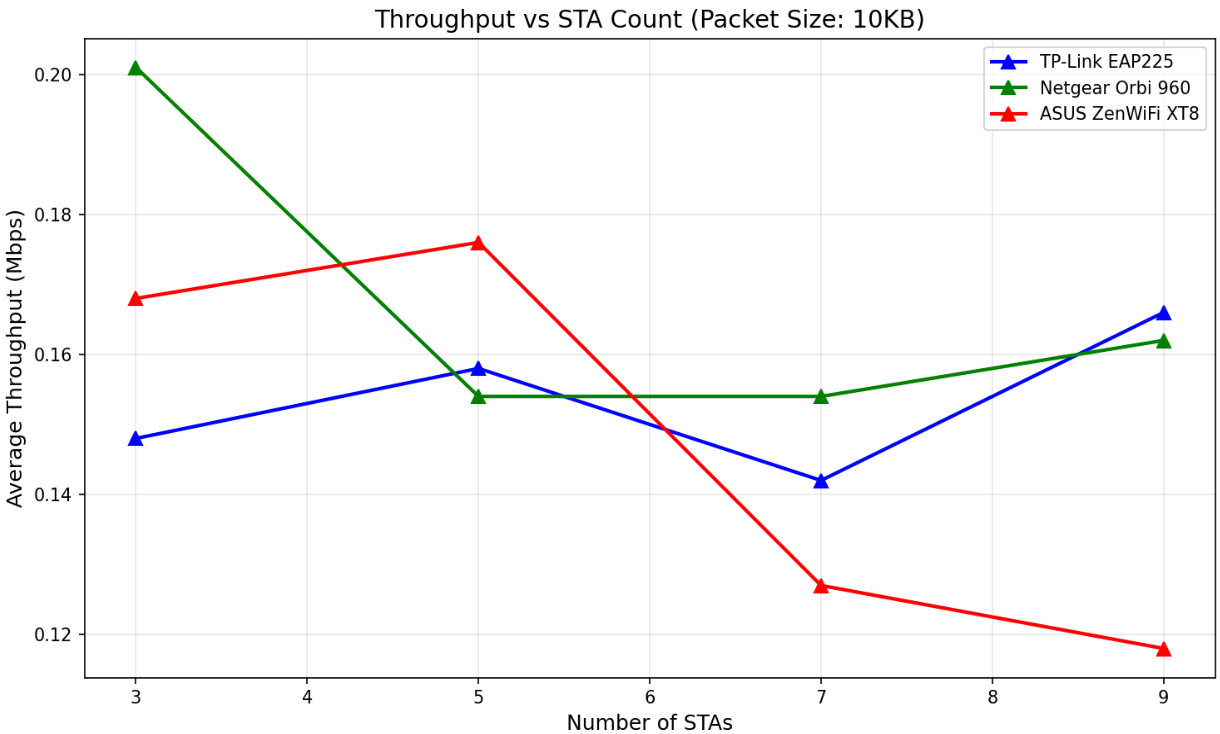
3. Performance Comparison

Config	Device	STAs	Avg Delay (ms)	PDR (%)	Throughput (Mbps)	Hops
1	TP-Link EAP225	3	14.24	91.8	0.15	0
1	TP-Link EAP225	5	28.11	88.8	0.16	0
1	TP-Link EAP225	7	30.03	88.2	0.14	0
1	TP-Link EAP225	9	38.16	86.3	0.17	0
2	Netgear Orbi 960	3	19.52	91.7	0.20	0
2	Netgear Orbi 960	5	24.31	89.4	0.15	0
2	Netgear Orbi 960	7	31.06	87.3	0.15	0
2	Netgear Orbi 960	9	48.84	88.9	0.16	0
3	ASUS ZenWiFi XT8	3	18.86	92.8	0.17	0
3	ASUS ZenWiFi XT8	5	30.94	90.2	0.18	0
3	ASUS ZenWiFi XT8	7	22.80	87.4	0.13	0
3	ASUS ZenWiFi XT8	9	38.26	81.1	0.12	0

4. Performance Analysis Graphs






Throughput vs STA Count (Packet Size: 10KB)
TP-Link EAP225
Netgear Orbi 960
ASUS ZenWiFi XT8
Average Throughput (Mbps)
0.20
0.18
0.16
0.14
0.12
3
4
5
6
7
8
9
Number of STAs

5. Mesh Path Analysis

Sample routing paths observed during tests:

Config	Device	STAs	Routing Path
1	TP-Link EAP225	3	
2	Netgear Orbi 960	3	
3	ASUS ZenWiFi XT8	3	

The routing paths show multi-hop mesh forwarding from the mobile STA (through AP Node 8) to the gateway (Node 0) and onward to the internet. Path length varies based on mesh configuration and network conditions. Buildings create obstacles that force traffic through alternate routes.

6. Conclusions

Based on the performance analysis with 10KB packets: • All three mesh configurations successfully delivered packets across the 400m × 400m area • Packet delivery ratio remained high across all tested STA counts • End-to-end delay increased with higher STA counts due to increased contention • The mesh network successfully routed around building obstacles Device Performance Summary: • TP-Link EAP225: Avg delay 27.64ms, Avg PDR 88.8% • Netgear Orbi 960: Avg delay 30.93ms, Avg PDR 89.3% • ASUS ZenWiFi XT8: Avg delay 27.71ms, Avg PDR 87.9%